

Exp No:3**Implementation of Parser using LEX and YACC tool****Aim**

To implement a **parser using LEX and YACC** that checks whether the given input string follows the specified grammar and determines whether the input is **valid or invalid**.

Algorithm

1. Start the program.
2. Create a Lex file to recognize identifiers and operators.
3. Create a YACC file containing grammar rules for valid expressions.
4. Save the Lex file as parser.l.
5. Save the YACC file as parser.y.
6. Open Command Prompt and navigate to the folder containing the files.
7. Execute flex parser.l to generate lex.yy.c.
8. Execute bison -dy parser.y to generate y.tab.c and y.tab.h.
9. Compile the generated C files using GCC.
10. Execute the generated parser.exe file.
11. Enter an expression.
12. YACC checks the expression according to the grammar rules.
13. If the expression follows the grammar, display **Valid Expression**.
14. Otherwise, display **Invalid Expression**.
15. Stop the program.

Lex Program — parser.l

```
%{
#include "y.tab.h"
%}

%%

[a-zA-Z][a-zA-Z0-9_]* { return ID; }
"+"                { return '+'; }
"*"                { return '*'; }
"("                { return '('; }
")"                { return ')'; }
"\n"               { return '\n'; }
[ \t]              { }

.                  { return yytext[0]; }

%%

int yywrap()
{
    return 1;
}
```

YACC Program — parser.y

```
%{
#include <stdio.h>

int yylex();
int yyerror(char *s);
%}

%token ID

%%

input:
    expr '\n'    { printf("Valid Expression\n"); }
    ;

expr:
    expr '+' term
    | term
    ;

term:
    term '*' factor
    | factor
    ;

factor:
    ID
    | '(' expr ')'
    ;

%%

int yyerror(char *s)
{
    printf("Invalid Expression\n");
    return 0;
}

int main()
{
    printf("Enter an expression: ");
    yyparse();
    return 0;
}
```

Procedure to Execute

Step 1: Create the LEX File

Open **Notepad** and paste the above Lex program.

Save it as:

parser.l

While saving, select:

Save as type → All Files

Make sure it is saved as parser.l and not parser.l.txt.

Step 2: Create the YACC File

Open another **Notepad** and paste the YACC program.

Save it as:

parser.y

Select:

Save as type → All Files

Make sure it is saved as parser.y.

Step 3: Open Command Prompt

Navigate to the folder containing the two files.

For example:

cd P:\pcd_lab\exp_3

Check the files:

ls

You should see:

parser.l

parser.y

Step 4: Generate Lexical Analyzer

Run:

flex parser.l

If there is no error, lex.yy.c will be generated.

Step 5: Generate YACC Parser

Run:

bison -dy parser.y

If there is no error, the following files will be generated:

y.tab.c

y.tab.h

Step 6: Compile

Run:

gcc lex.yy.c y.tab.c -o parser.exe

If there is no error, parser.exe will be created.

Step 7: Run the Parser

Run:

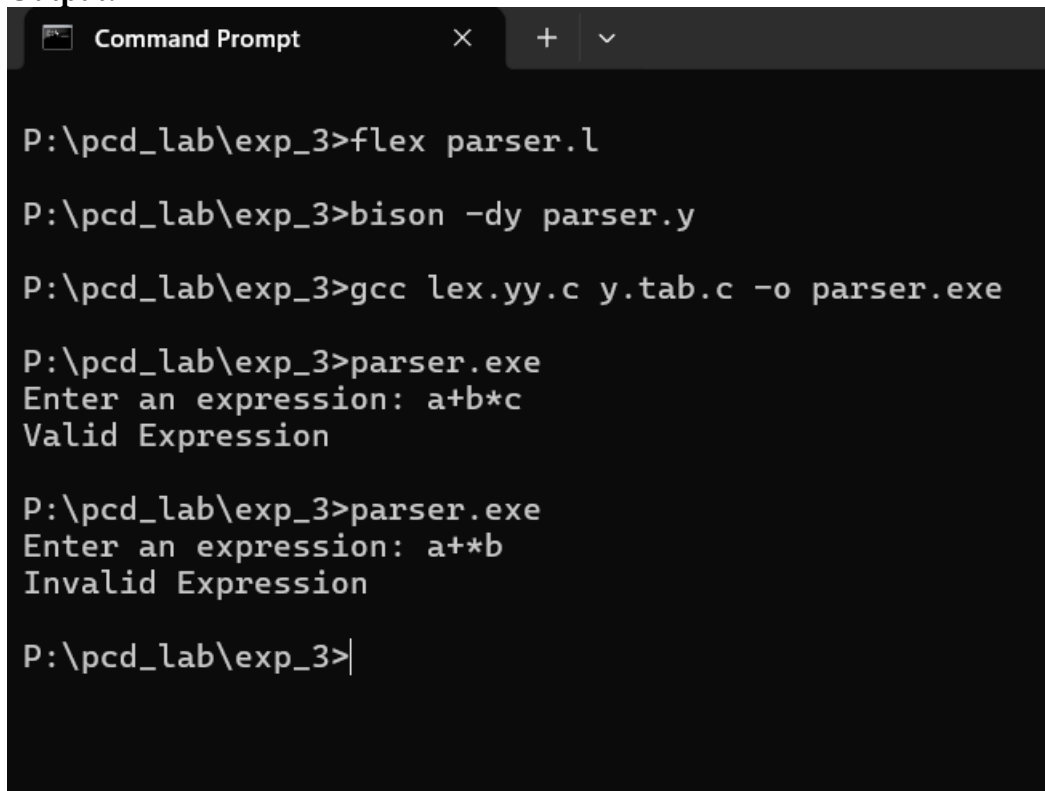
parser.exe

The program will ask:

Enter an expression:

Enter:
a+b*c
Press **Enter**.

Output:



```
Command Prompt
P:\pcd_lab\exp_3>flex parser.l
P:\pcd_lab\exp_3>bison -dy parser.y
P:\pcd_lab\exp_3>gcc lex.yy.c y.tab.c -o parser.exe
P:\pcd_lab\exp_3>parser.exe
Enter an expression: a+b*c
Valid Expression

P:\pcd_lab\exp_3>parser.exe
Enter an expression: a+*b
Invalid Expression

P:\pcd_lab\exp_3>|
```

Result:

Thus, a **parser** was **successfully implemented using LEX and YACC**, and the parser successfully identified whether the given input expression was valid or invalid according to the specified grammar.